Simulation Output Summary

Integrated Energy Systems Optimiser (IESO) v24.03 · www.github.com/greoux-research/ieso

March 25, 2024 09:15 AM

Demand

Demand for Electricity:

- Annual Demand in MWh: 50000000
- Hourly Profile: Figure 1
- Non-Service Penalty in $ per MWh: 20000
- Supply Sources: Power Grid

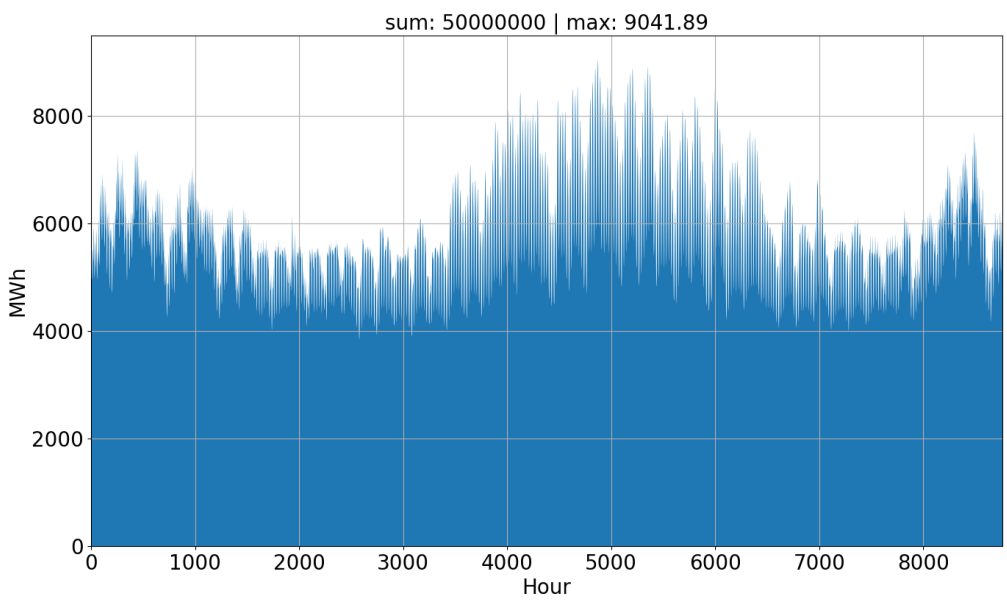

Figure 1: Demand for Electricity

Demand for Water:

- Annual Demand in Q(X): 365000000

- Hourly Profile: Uniform (41667 Q(X) each Hour)
- Non-Service Penalty in $ per Q(X): 1000
- Supply Sources: Power-to-X Processes ["p2x-ro"]

Generators

Generator "nucl":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 456466
- Variable Costs in $ per MWh: 11.5
- Variable Emissions in $ per MWh: 0
- The optimiser did not suggest adding generation capacity

Generator "coal":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 222392
- Variable Costs in $ per MWh: 26.84
- Variable Emissions in $ per MWh: 880
- Capacity in MW: 2342.73
- Electricity Generation: 13156535 MWh (Figure 2)
- Number of Successive Changes in Output: 2706

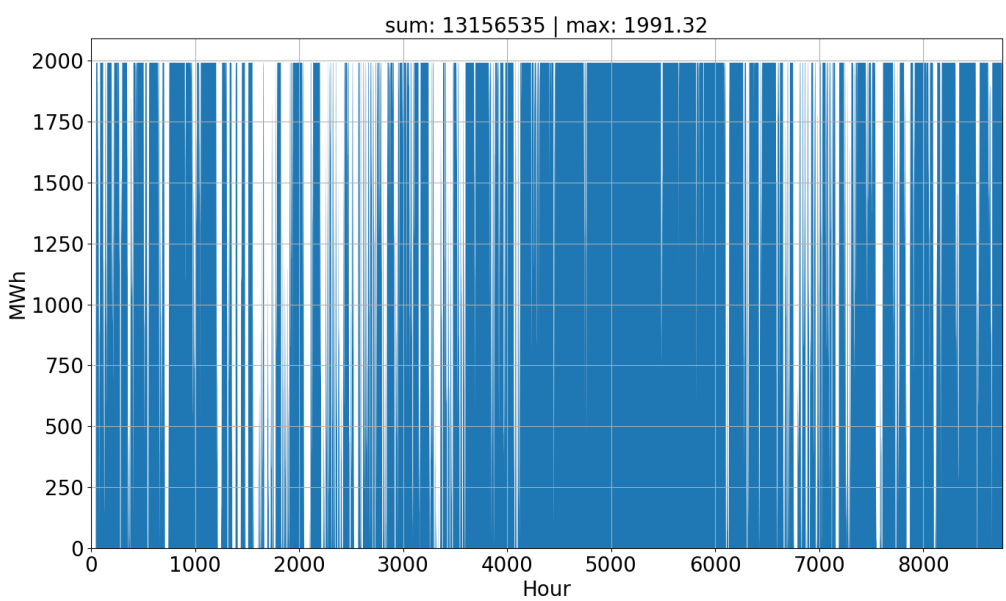

Figure 2: Electricity Generation by "coal"

Generator "ccgt":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 98749
- Variable Costs in $ per MWh: 56.44
- Variable Emissions in $ per MWh: 365
- Capacity in MW: 3001.14
- Electricity Generation: 6687374 MWh (Figure 3)
- Number of Successive Changes in Output: 3831

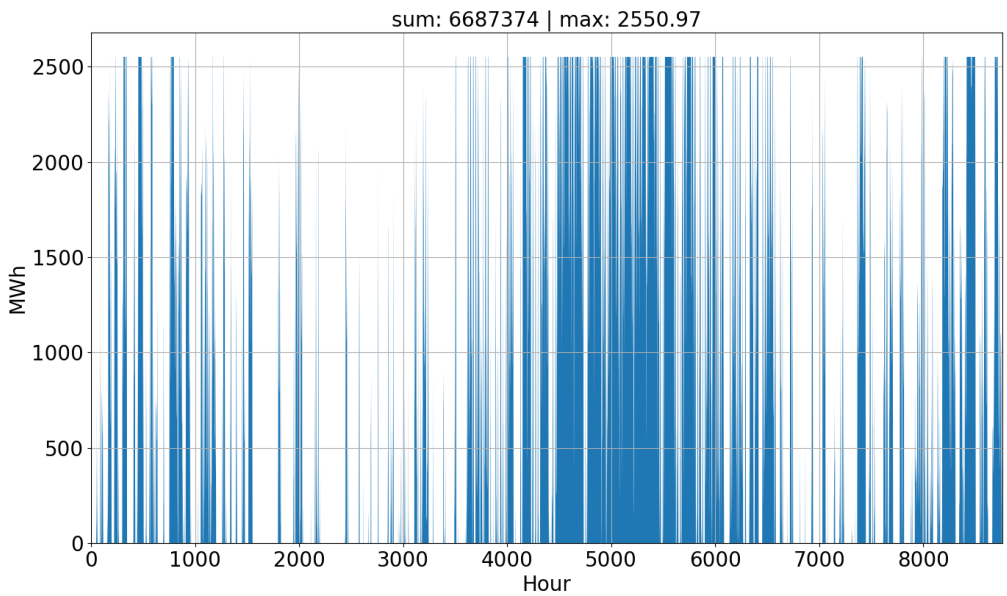

Figure 3: Electricity Generation by "ccgt"

Generator "ocgt":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 75182
- Variable Costs in $ per MWh: 96.11
- Variable Emissions in $ per MWh: 523
- Capacity in MW: 2049.59
- Electricity Generation: 372682 MWh (Figure 4)
- Number of Successive Changes in Output: 620

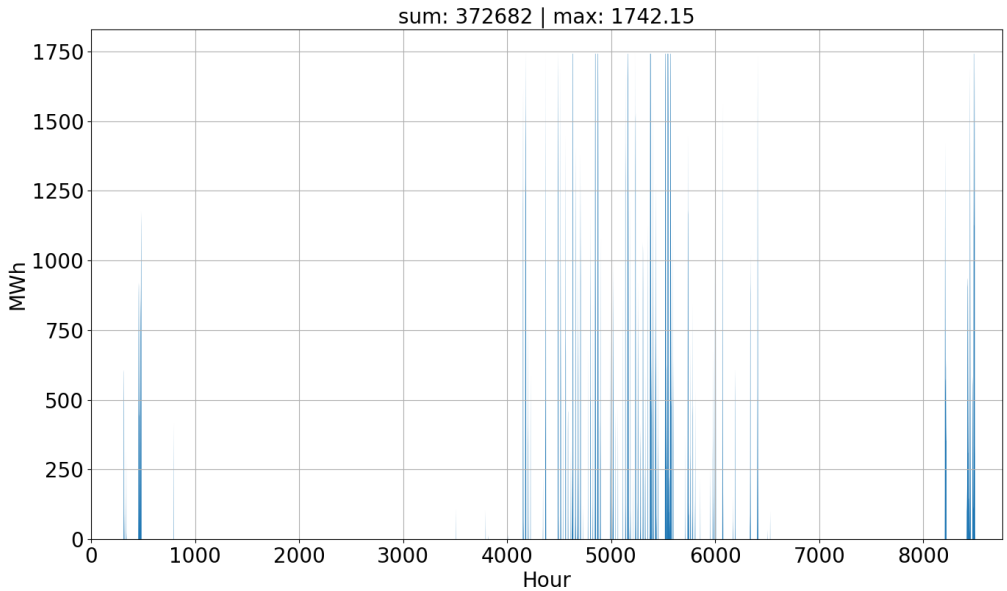

Figure 4: Electricity Generation by "ocgt"

Generator "solr":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 70046
- Variable Costs in $ per MWh: 3e-06
- Variable Emissions in $ per MWh: 0
- Capacity in MW: 5089.93
- Electricity Generation: 8367777 MWh (Figure 5)
- Curtailment: 0 Hours; 0 MWh or 0.0 % of the Tech.'s Annual Gen. Volume

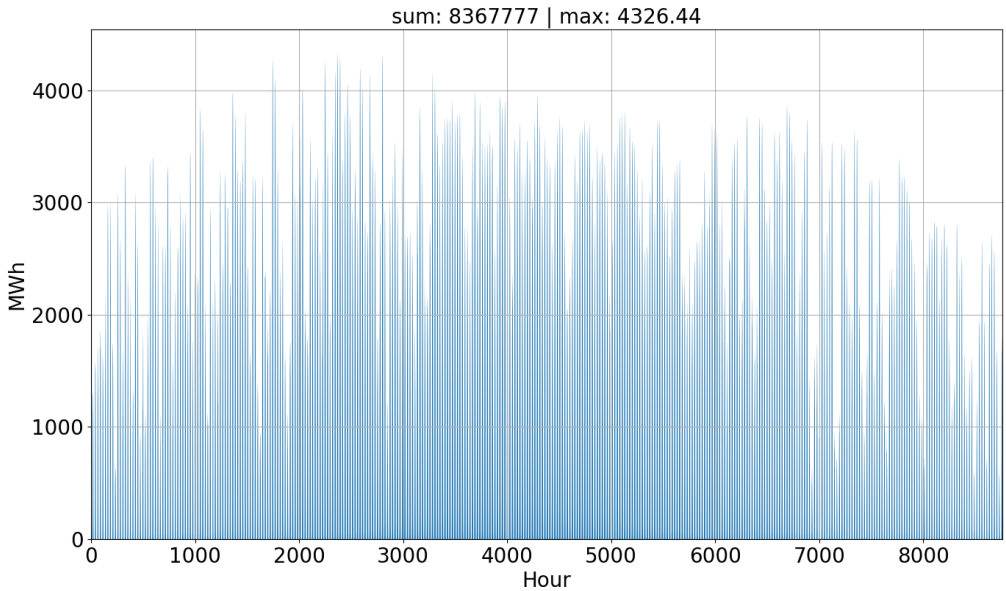

Figure 5: Electricity Generation by "solr"

Generator "wind":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 133553
- Variable Costs in $ per MWh: 6e-06
- Variable Emissions in $ per MWh: 0
- Capacity in MW: 6905.74
- Electricity Generation: 23188247 MWh (Figure 6)
- Curtailment: 580 Hours; 551671.64 MWh or 2.38 % of the Tech.'s Annual Gen. Volume

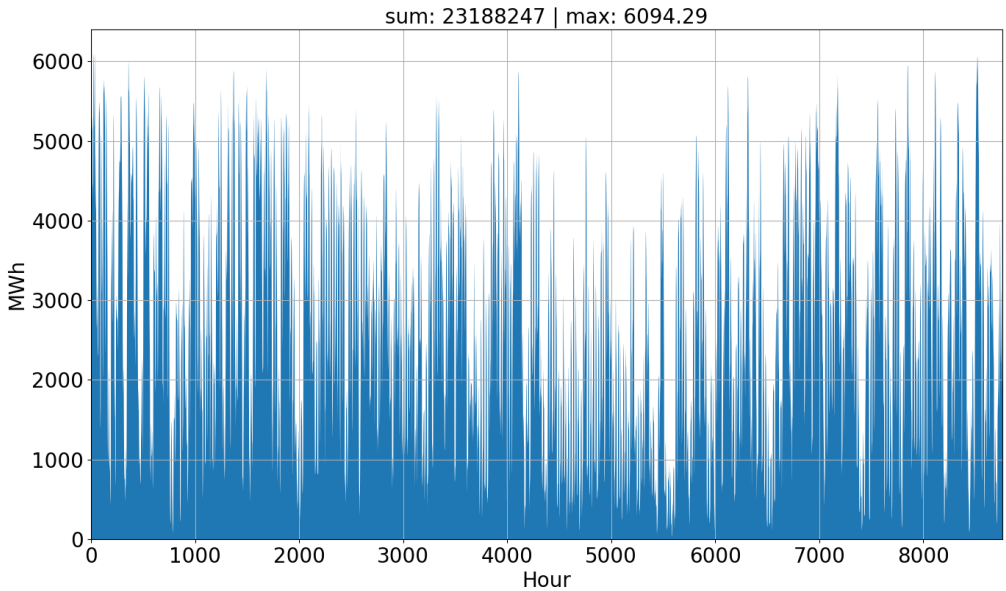

Figure 6: Electricity Generation by "wind"

Flexibility Means

Flexibility Mean "bstr":

- Type: Electricity Storage System
- Fixed Costs in $ per MWh per Year: 27248
- Hours of Storage at Maximum Discharge: 4
- Round Trip Efficiency: 0.85
- Capacity in MWh: 3689.71
- Electricity Charged (-) and Discharged (+): Figure 7
- Electricity Stored: Figure 8

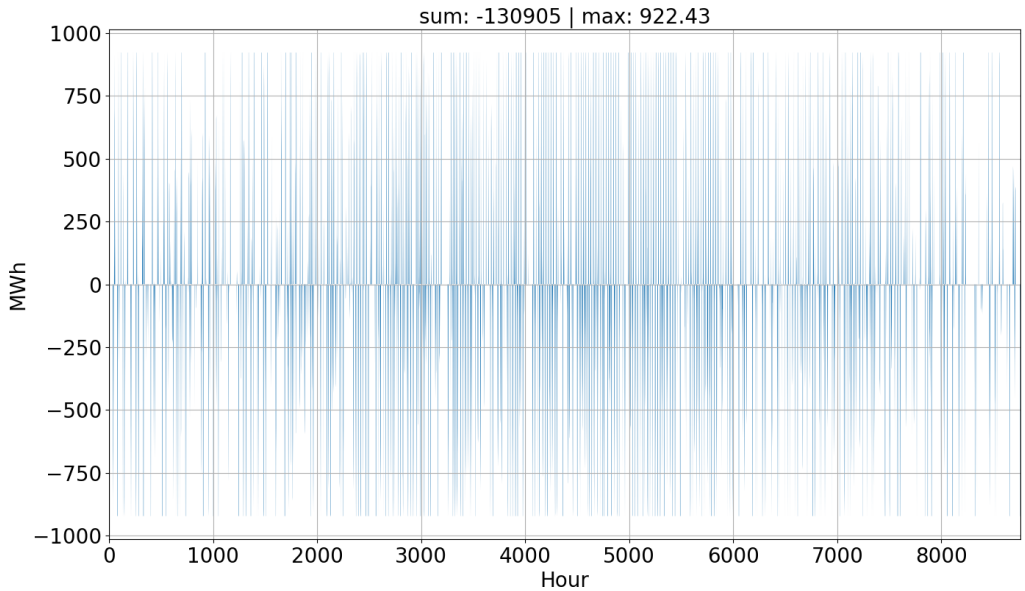

Figure 7: Electricity Charged (-) and Discharged (+) by "bstr"

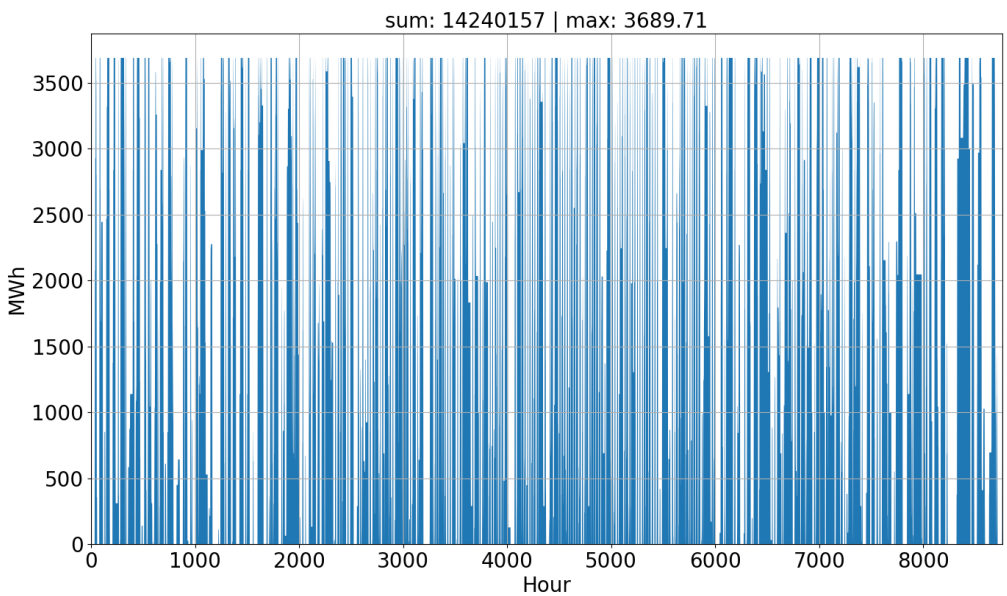

Figure 8: Electricity Stored by "bstr"

Power-to-X Processes

Power-to-X Process "p2x-ro":

- Consumes Electricity (Grid-Supplied 0.0045 MWh per Q(X))
- Fixed Production Costs in $ per Q(X) per Hour per Year: 2070
- Variable Production Costs (Excluding Energy) in $ per Q(X): 0.35
- Production Capacity in Q(X) per Hour: 49096.77
- Production Profile: Figure 9
- Fixed Storage Costs in $ per Q(X) per Year: 0.22
- Storage Capacity: 361930.34 Q(X) or 9 Hours of Production at Max. Capacity
- Storage Profile: Figure 10
- Supply Profile: Figure 11

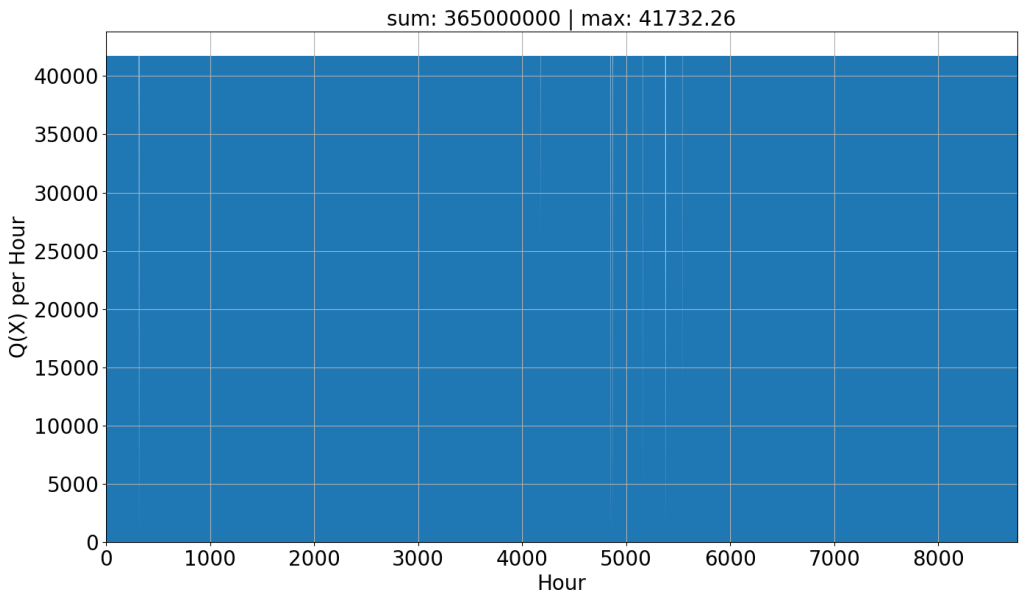

Figure 9: Production Profile - "p2x-ro"

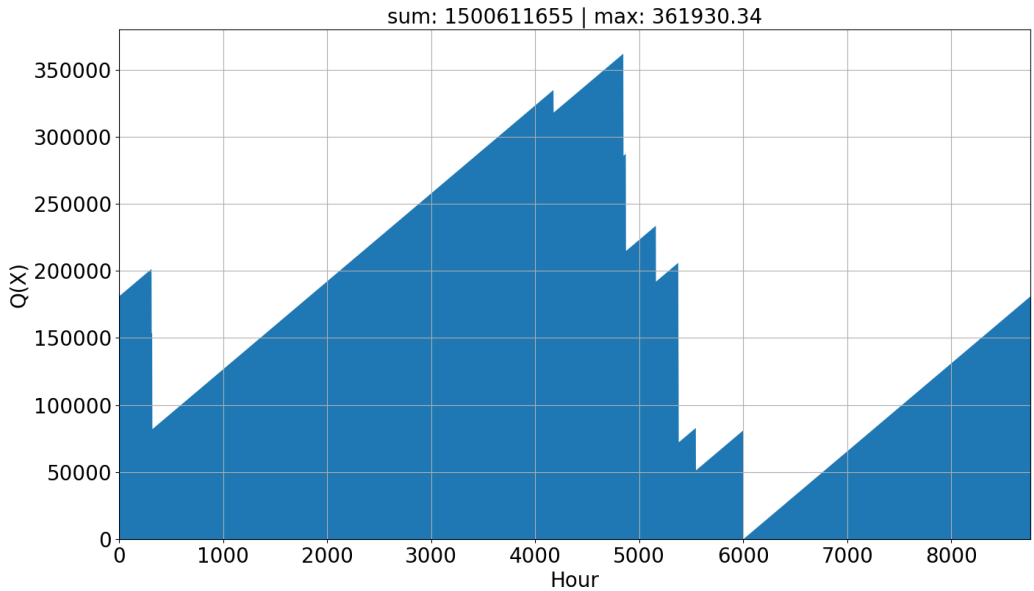

Figure 10: Storage Profile - "p2x-ro"

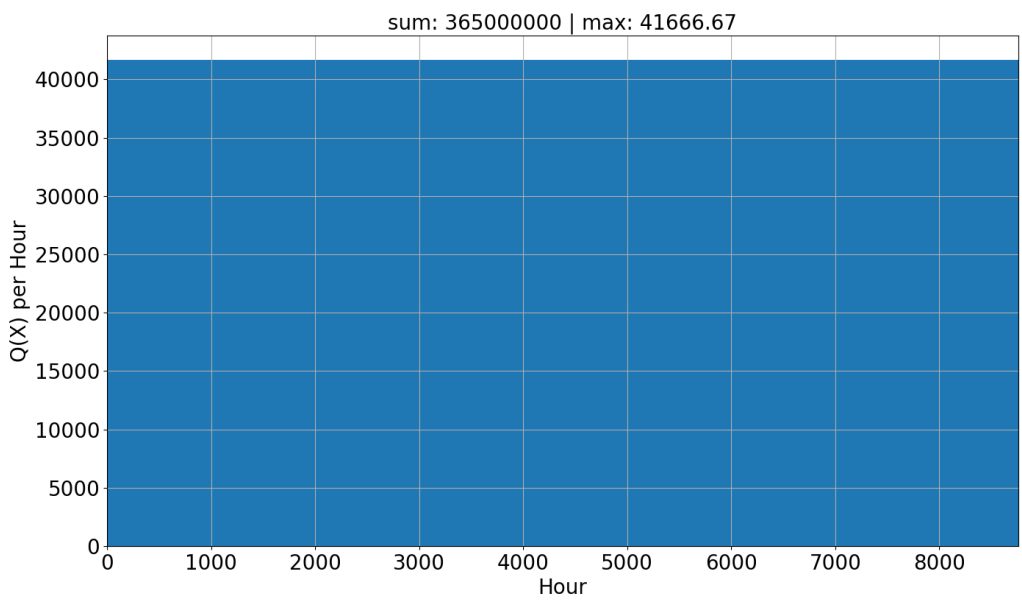

Figure 11: Supply Profile - "p2x-ro"

Costs and Emissions

Generation Costs and Emissions on a per MWh Basis:

- Generation Costs in $ per MWh: 58.27
- Generation Emissions in kg per MWh: 274.54

Prices

Electricity:

- Average Price in $ per MWh: 61.21
- Price Profile: Figure 12

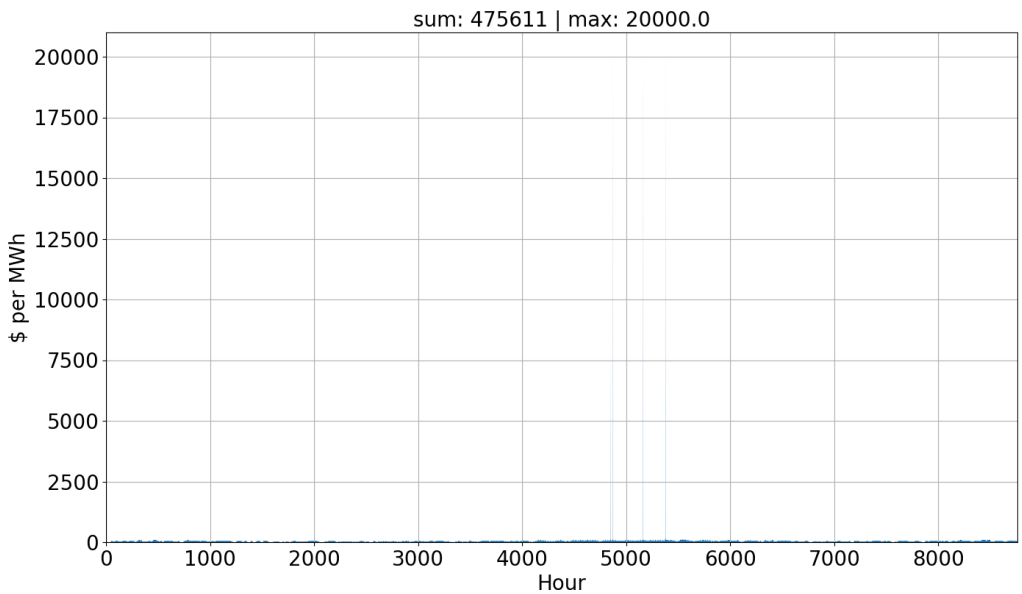

Figure 12: Electricity Price Profile

Water:

- Average Price in $ per Q(X): 0.83
- Price Profile: Figure 13

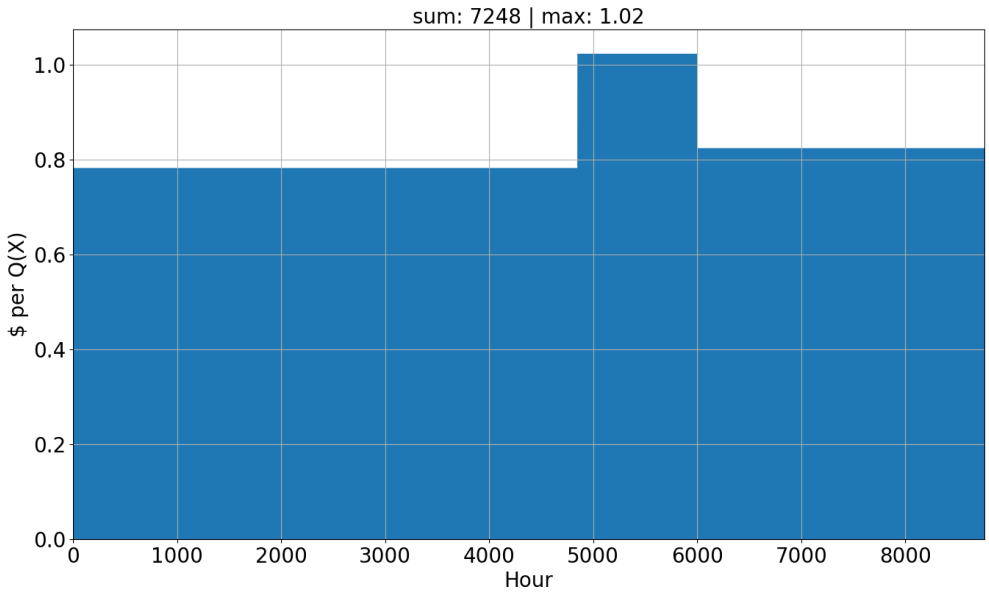

Figure 13: Water Price Profile

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